22 There are only red counters, blue counters and green counters in a bag.

number of red counters : number of blue counters : number of green counters = 1 : 3 : 7

A counter is going to be taken at random from the bag.

(a) Complete the table below to show each of the probabilities that the counter will be red or blue or green.

1 mark
Total11

Colour	red	blue	green
Probability	$\frac{1}{11}$	$\frac{3}{11}$	$\frac{7}{11}$

1 mark
All three fractions

1 + 3 + 7 = 11 total parts.

(2)

Jamie takes at random a counter from the bag and records the colour of the counter.
He then puts the counter back in the bag.

Jamie does this a number of times.
He records a total of 68 blue counters.

(b) Work out an estimate for the total number of times Jamie takes a counter from the bag.

Blue	Total
3	11

68 blue represents3 parts.

1 mark
Divide by3/11

Estimated total = 68 × 11 ÷ 3 = 249.3333...

About249 selections.

1 mark
249 (or250)

(2)

(Total for Question 22 is 4 marks)